Anonymization as a Group Action:

Zero-Leakage Timbre Replacement for Clinical Speech, and the Limits of Parameter-Domain Invariance

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Abstract

Speech anonymization for clinical corpora is constrained by three competing requirements: resistance to automatic speaker verification, retention of lexical content, and preservation of the acoustic biomarkers that make the recording clinically useful. We present a system that closes the identity channel with a formal information-theoretic guarantee and measure, rather than assume, what that costs on the other two axes.

The mechanism, ORBIT-N+F, has three stages. *Stage A (PolarQuant)* replaces the speaker's long-term timbre with a draw from a public population: in whitened population coordinates a keyed Haar rotation maps the speaker cloud onto itself, b directional sign bits are retained, and the direction is resampled rather than reconstructed. At $b = 0$ the data-processing inequality gives $I(\text{timbre}_{\text{in}}; \text{timbre}_{\text{out}}) = 0$ bits—independence, which we confirm empirically (whitened cosine $+0.017 \pm 0.081$ over 15 releases). *Stage B* releases the vocal-tract scale under Gaussian differential privacy against public anchors derived without reading the protected corpus. *Stage C* dilates only the liftered spectral envelope, leaving the harmonic comb—and therefore pitch—in place (measured drift ≤ 0.002 st).

On a 14-identity Swedish clinical corpus, scored throughout by one evaluator, the strongest configuration (ORBIT-N+F → OpenVoice V2) reaches **EER 40.3%**, **self-cosine −4.3%**, $\boldsymbol{\pi_{\text{link}} = 0.071}$—exactly chance for $N = 14$—and **min C_{llr} 0.799**. A control in which the converter converts each speaker into themselves leaves identity intact ($\pi_{\text{link}} = 1.000$), establishing that the privacy is the mechanism's and not the vocoder's.

The cost is substantial and we report it: **CER 34.1%** against 14.6% for ORBIT-N+F's own renderer, and **ICC(2,1) between −0.32 and 0.31** across nine clinical markers, none above the 0.75 acceptability threshold. A 2×2 factorial attributes **+15.5 CER points to the timbre replacement and +5.2 to the dilation, with an interaction of −1.2**—additive, which refutes the "double spectral distortion" explanation we began with. We therefore do not claim the trilemma is resolved. We claim it is characterized: the configuration with the strongest unlinkability is not the one with the best fidelity, and on this evidence **ORBIT-N+F alone remains the better system for clinical use**.

Keywords: speech anonymization; voice privacy; Gaussian differential privacy; voice conversion; invariant theory; clinical speech biomarkers.

1 Introduction

1.1 The trilemma

A speech anonymizer for clinical data is judged on three axes that do not move together (Table 1).

The tension is structural. Identity in speech is carried redundantly—by the long-term spectral envelope, by vocal-tract scale, by pitch register, by micro-prosody—and the channels that carry it are the same channels that carry vowel identity, articulation rate and voice quality.

Table 1: The three evaluation axes.

Axis	Metric	Direction
Machine security	EER of a speaker-verification attack; π_{link}; min C_{llr}	higher / lower / higher
Intelligibility	WER, CER of a recognizer	lower
Clinical fidelity	ICC(2,1) of acoustic biomarkers	higher

Removing identity without touching content is not a tuning problem; it requires finding a channel that carries identity and *not* the clinical signal, and establishing that the chosen channel is genuinely separable.

1.2 Limitations of existing approaches, as measured here

Classical DSP. McAdams coefficient warping [15] and vocal-tract length normalisation (VTLN) [8] perturb the speaker's own parameters. The output remains a function of the input: the transform is a bijection on a low-dimensional parameter, so identity entropy is conserved and a parameter sweep inverts it. Measured on our corpus, McAdams at $\alpha = 0.92$ leaves **57.6% self-cosine** and VTLN-P with per-speaker jitter leaves **39.8%**—more than a third of the embedding survives.

Unconstrained neural voice conversion. A converter that re-renders the waveform from the speaker's *own* conditioning vector changes the signal substantially without changing the identity. Our control run—OpenVoice V2 [16] converting each speaker into themselves—scores $\boldsymbol{\pi_{\text{link}} = 1.000}$: all 14 speakers are still correctly linked. This control is load-bearing for everything that follows, because a verification embedding cannot distinguish "this is someone else" from "this has been through a vocoder."

Cascaded DSP + neural VC. We began with the hypothesis that cascading a spectral pre-warp into a converter produces compounding distortion. Section 7.1 reports the factorial that refutes it.

1.3 Contributions

1. A timbre-release mechanism with a **zero-bit information cap**, verified as independence rather than as small similarity (§ 3.2).
2. A **μ-GDP release of vocal-tract scale** whose anchors are derived from public speech, with the commensurability of the released coordinate established by measurement (§ 3.3.2).
3. A **pitch-preserving envelope dilation** rendered back to a waveform without a second generative model (§ 3.4).
4. A **measured account of the costs**, including negative results that constrain the design space: the stages are additive, not compounding (§ 7.1); the embedding-regime hypothesis does not explain the cost (§ 7.2); and the measured intelligibility cost is strongly recognizer-dependent (§ 7.3).

2 Threat model and evaluation protocol

2.1 Attackers

Following the convention of the evaluator used throughout, three scenarios are scored:

- **Ignorant** ($v_0 \to v_2$): the attacker enrols on original recordings and tests against releases. This is the headline EER.
- **Lazy-informed** ($v_2 \to v_2$): the attacker enrols and tests on releases. This measures whether speakers remain mutually distinguishable—which diarization needs, and which is not a privacy failure against an attacker holding original audio.
- **Baseline** ($v_0 \to v_0$): untouched audio, establishing the attack's own floor. **A result is uninterpretable without it**; see § 7.4.

Embeddings are Resemblyzer d-vectors [22] over 3 s chunks, z-scored across the pooled corpus. Legal-style metrics—π_{link} (probability the correct speaker is the nearest enrolled one), π_{sing}, and min C_{llr} [2]—are computed from one mean embedding per identity with the remaining identities as a calibration set.

2.2 Keying

Every draw is keyed by SHA256(secret ‖ item ‖ speaker ‖ session). The session tag re-keys all draws, so one speaker in two recordings receives independent releases. Privacy loss across N recordings composes as approximately $\sqrt{N}\mu$ and is **not** tracked; diarizer labels are per-file.

2.3 Corpus

Seven Swedish clinical consultations, 49.1 minutes, 14 scorable identities, 851 chunks. Intelligibility is scored by `KBLab/wav2vec2-large-voxrex-swedish` (CTC) [12] with the reference being the recognizer's own transcript of the **original**, so the measure charges the anonymizer only for damage it caused. Section 7.3 shows this choice is consequential.

3 System

3.1 Architecture

Figure 1 shows the signal flow. Analysis uses the WORLD vocoder [14] (Harvest → StoneMask → CheapTrick → D4C) together with an 80-band mel spectrogram at 24 kHz ($n_{\text{fft}} = 1920$, hop 480). Pitch is **not** an output of any stage. It passes through, and § 8 states the consequence.

3.2 Stage A — PolarQuant timbre release

3.2.1 Construction

Let $\bar{m}(f)$ be the gain-removed mean log spectral envelope over speech frames. Take

$$c = \text{DCT-II}(\bar{m})[1..K_{\text{hi}}], \qquad K_{\text{hi}} = 48, \tag{1}$$

which holds **99.6%** of between-speaker mean-shape variance on the fitted population, against 88.2% for $K = 12$; the residual was distinct per speaker and functioned as a fingerprint.

A population model fitted on **329 public RixVox speakers** [17] supplies a whitening. Writing $\tau = \text{to_tau}(c, \bar{\phi})$ for the whitened coordinate (conditioned on mean log-F_0, so a released register is paired with a plausible timbre), the population is isotropic in τ, and the polar split is

$$\tau = r \cdot u, \qquad r = \|\tau\|_2, \quad u = \tau / r. \tag{2}$$

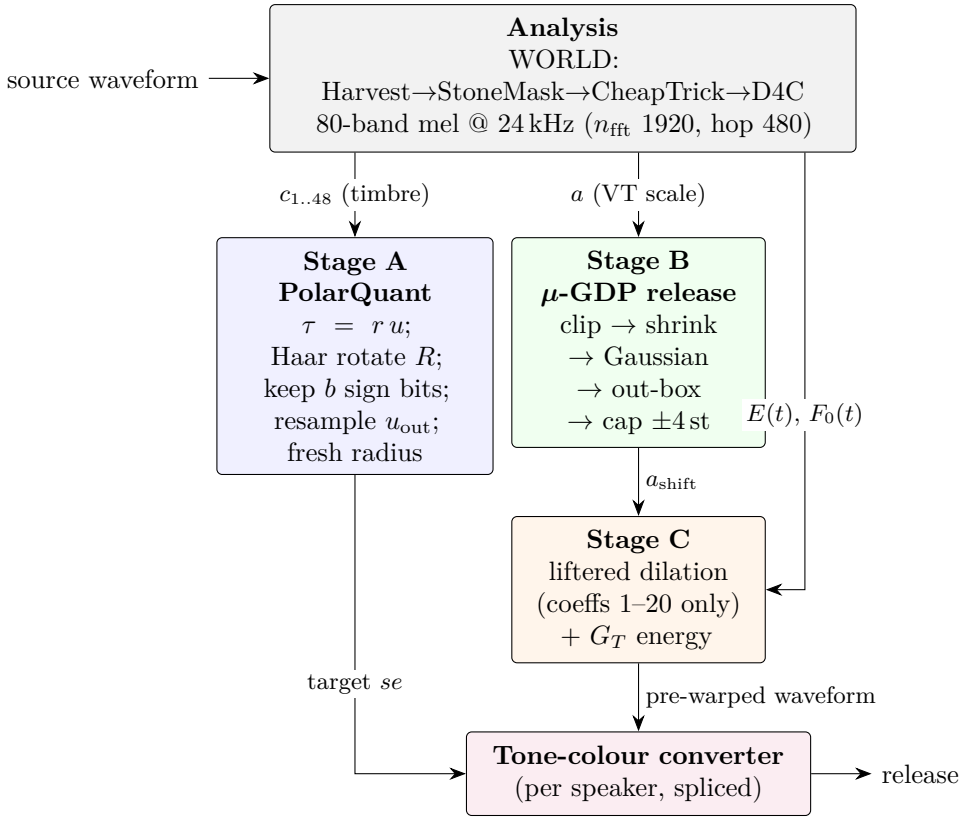


Figure 1: ORBIT-N+F with a neural voice-conversion back end. Stage A releases timbre, Stage B releases vocal-tract scale, Stage C renders a pitch-preserving dilation. F_0 passes through untouched.

Per speaker per session, keyed:

$$R \sim \text{Haar}(K_{\text{hi}}), \qquad v = Ru, \qquad s_i = \text{sign}(v_i) \quad (i \le b), \tag{3}$$

$$g \sim \mathcal{N}(0, \mathbf{I}), \quad g_i \leftarrow |g_i|\, s_i \quad (i \le b), \qquad u_{\text{out}} = R^{\mathsf{T}}\left(\frac{g}{\|g\|}\right), \tag{4}$$

$$r_{\text{out}} = \min\big(\chi_{K_{\text{hi}}} \text{ draw},\, q_{95}\big), \qquad \tau_{\text{out}} = r_{\text{out}}\, u_{\text{out}}. \tag{5}$$

Two choices are load-bearing. **The rotation is Haar** [13] because in whitened coordinates the population is isotropic, so R maps it onto itself: a rotated speaker is exactly as plausible as the original. Perturbation methods such as jitter push a sample *off* the manifold of real voices, which is what sounds synthetic; rotation moves along it. **The direction is resampled, not reconstructed at a cell centroid**, because centroid reconstruction averages voices together and produces the characterless timbre typical of quantized speech.

3.2.2 Zero-leakage theorem

Proposition 1. *The released direction depends on the input only through the b retained sign bits. Hence, by the data-processing inequality [4],*

$$I(\textit{timbre}_{in}; \textit{timbre}_{out}) \le b \text{ bits}, \tag{6}$$

and at $b = 0$, $I(\textit{timbre}_{in}; \textit{timbre}_{out}) = 0$ bits.

Proof. The Markov chain is $\tau_{\text{in}} \to S_b \to \tau_{\text{out}}$ where $S_b = (s_1, \ldots, s_b)$. Given R (independent of the input) and S_b, the draw g and hence u_{out} are conditionally independent of τ_{in}; r_{out} is drawn from the population law without reference to τ_{in}. The DPI gives $I(\tau_{\text{in}}; \tau_{\text{out}}) \le I(\tau_{\text{in}}; S_b) \le H(S_b) \le b$. At $b = 0$, S_b is empty and $\tau_{\text{out}} \perp \tau_{\text{in}}$. □

With `--polar-flip p`, retained bits receive randomized response [23], making the release $b \ln \frac{1-p}{p}$-DP [7]—a formal statement rather than an information cap.

Empirical confirmation. Independence, not merely low similarity, is the claim, and it is testable: the same key must produce the same target whatever speaker is presented. Over 15 speaker-releases we measure the quantities in Table 2.

Table 2: Input–output similarity of the released timbre over 15 speaker-releases ($b = 0$).

Quantity	mean ± sd
whitened cosine $\cos(\tau_{\text{in}}, \tau_{\text{out}})$	$\mathbf{+0.017 \pm 0.081}$
centred latent cosine	$+0.028 \pm 0.201$
raw 256-d embedding cosine	$+0.454 \pm 0.138$

The raw cosine is **not** residual leakage: an un-centred embedding space has a large population mean that inflates every pairwise cosine. Reporting it alone would misstate the result in either direction, which is why all three are reported.

3.2.3 Reachability in a neural latent

When the back end is a neural converter, the target must be a vector the converter can actually receive. OpenVoice V2's tone-colour vector is the output of `ReferenceEncoder.proj`, an $\mathbb{R}^{128} \to \mathbb{R}^{256}$ affine map:

$$se = Wh + b, \qquad W \in \mathbb{R}^{256\times 128}. \tag{7}$$

We verify $\operatorname{rank}(W) = 128$ with condition number 4.9, and that $h = W^{+}(se - b)$ reconstructs 5000 public embeddings to $\mathbf{5.1 \times 10^{-8}}$ relative error. The reachable set is therefore a **128-dimensional affine subspace of $\mathbb{R}^{256}$**, not $\mathbb{R}^{256}$. Moreover h is a GRU hidden state, so $h \in (-1, 1)^{128}$ by induction, and all 5000 public vectors satisfy it.

Sampling in the full 256-d space would produce a conditioning vector no recording can yield. The release is therefore drawn in h, projected onto the box, and mapped out through W. The box projection reads only the draw, never the input, so it is post-processing and the information cap is untouched.

3.3 Stage B — μ-GDP release of vocal-tract scale

3.3.1 Mechanism

The released coordinate is a, the energy-weighted log-frequency centroid of the mean shape over $[150\,\text{Hz},\ 0.95 f_s/2]$:

$$a_c = \operatorname{clip}(a_{\text{in}},\ \text{anchor} \pm \Delta/2), \qquad a_{\text{out}} = \text{anchor} + \lambda(a_c - \text{anchor}) + \sigma z, \quad z \sim \mathcal{N}(0,1). \tag{8}$$

Clipping bounds the sensitivity; the Gaussian mechanism then gives $\mu = \lambda\Delta/\sigma$ under Gaussian differential privacy [6]. Noise on the scale and pitch coordinates is ρ-correlated, because a shorter vocal tract co-occurs with a higher voice in real speakers, and drawing them independently produced implausible combinations. Under correlated noise the sensitivity is measured in the Mahalanobis metric of Σ, giving in closed form

$$t = \mu\sqrt{\tfrac{1-|\rho|}{2}}, \qquad \lambda_j = t\,\sigma_j/\Delta_j. \tag{9}$$

Projecting the **released** coordinate into a plausibility box is post-processing and μ survives. Capping the **applied shift** at ± 4 st is not—it reads the private input—and every file where it fires is flagged `gdp_exact: false`. At the reference configuration $\lambda_a = 0.054$ and $\mu_a = 0.354$; across 15 releases the cap **never fired** (15/15 `gdp_exact: true`), against 5 of 45 in the predecessor system.

Anchors are public. The anchor (288.6 Hz) and clip width ($\Delta = 0.324$) are obtained by pushing 4000 seeded draws from the public speaker space through the same centroid functional. This matters: corpus-calibrated anchors enter every release un-noised at weight $1 - \lambda$ and also set λ, making them a function of the records being protected.

3.3.2 Commensurability of the released coordinate

A released coordinate is meaningless unless it is measured on the same object as the anchor. The anchors were computed on WORLD CheapTrick envelopes; the converter environment has no `pyworld`, so the coordinate is measured from an STFT. We treated this as a measurement to be made, not a detail to be assumed.

At the dilation's own 80 ms analysis window the harmonic comb is fully *resolved*, and the deep inter-harmonic valleys—which no envelope estimator ever sees—bias the centroid upward, by more at higher F_0. Against a clip window of ± 2.8 st this is a large systematic error. CheapTrick analyses pitch-synchronously over $3/F_0$ seconds (13–32 ms), so the coordinate is measured on its own short window (Table 3).

Table 3: Bias of the STFT centroid relative to the WORLD reference, by analysis window.

n_{fft}	256	384	512	768	1024	1920
bias median (st)	**−0.13**	−0.36	−0.39	−0.69	−0.43	**+2.43**
sd (st)	0.27	0.25	0.29	0.37	0.36	1.13

Final agreement with the WORLD reference at $n_{\text{fft}} = 256$: $\mathbf{-0.13 \pm 0.28}$ **st** on synthetic vowels spanning F_0 95–230 Hz, and $\mathbf{-0.34 \pm 0.32}$ **st** on the clinical corpus.

3.4 Stage C — liftered dilation and G_T

3.4.1 Invariance in the parameter domain

Let $G_T = \{c \in \mathbb{R}^F\}$ act on the log envelope by $L(t, f) \mapsto L(t, f) + c(f)$ followed by exact per-frame energy restoration. Theorem 3 (§ 4.6) shows that the time-varying spectral residual D is fixed identically by this action. Consequently `spec_var_db` and `artic_dyn_db_s`, which are functionals of D alone, are invariant; `pause_ratio`, `shimmer_rel` and `loud_var_db` are functionals of frame energy $E(t)$, restored exactly; and the ψ-markers (`f0_var_st`, `f0_range_st`, `jitter_rel`) never see the envelope. All nine are invariant, with worst-case drift $2.7 \times 10^{-14}\%$—floating-point precision.

This is a parameter-domain statement. Section 6.3 reports what survives end-to-end, and the answer is: little, once a generative decoder is downstream. We regard stating this plainly as part of the contribution.

3.4.2 Pitch-preserving dilation

A log-mel frame splits by cepstral liftering into a smooth envelope (first 20 DCT coefficients across bands: formants, tilt) and fine structure (harmonic ripple, where pitch lives). Only the envelope is warped by $\exp(-a\, w(f))$, tapered to identity over the top octave; the fine structure is added back unmoved. Warping the whole mel would move the harmonics and shift pitch by the same ratio.

The operator's effect is $\delta(t, b) = \text{env}_{\text{warped}} - \text{env}$—a difference of two 20-coefficient curves, hence smooth in frequency. A smooth log-magnitude gain moves no harmonic, so it can be applied to a linear STFT and inverted with the **original phase**, avoiding a second generative model entirely.

Measured: a formant at 708 Hz lands at 894 Hz for a requested +4 st (**+4.04** measured) and at 559 Hz for −4 st (**−4.09**). Median pitch drift is $\leq$ **0.002 st** on every corpus file against a 0.05 st budget.

4 ORBIT-N+F: deriving the mechanism from an invariance requirement

Sections 3.2 to 3.4 described *what* the stages do. This section gives the argument that produced them. The design is not an assembly of perturbations chosen for effect; it is a group action selected so that the set of surviving clinical statistics is **characterized in advance** rather than discovered by trial.

4.1 Anonymization as a group action

Write the signal in the coordinates where identity acts linearly:

$$u = \log f, \qquad \phi(t) = 12 \log_2\big(F_0(t)/f_{\text{ref}}\big). \tag{10}$$

In these coordinates the two dominant speaker attributes are **translations**:

$$\text{vocal-tract scale } \alpha: \quad S(u,t) \mapsto S(u - \log\alpha,\, t), \tag{11}$$

$$\text{pitch transposition } \beta: \quad \phi(t) \mapsto \phi(t) + \beta. \tag{12}$$

This is the pivotal modelling choice. Translation groups possess a **complete invariant theory**, so the question "which clinical statistics survive?" has an exact answer rather than an empirical one.

4.2 Maximal invariant

Theorem 1 (Maximal invariant)**.** *For $G = \{\alpha\} \times \{\beta\}$ the pair*

$$\psi(t) = \phi(t) - \operatorname*{mean}_t \phi(t), \qquad M(t,\cdot) = \big|\mathcal{F}_u\{\log S(u,t)\}\big| \tag{13}$$

*is a maximal invariant for the action. A statistic T is G-invariant **if and only if** it factors through (ψ, M).*

ψ is the mean-removed log-F_0 contour: a translation $\phi \mapsto \phi + \beta$ shifts the mean by β and leaves ψ fixed. M is the scale-transform (Mellin) magnitude: a translation in u multiplies $\mathcal{F}_u$ by a unit-modulus phase, which the magnitude discards.

The force of the theorem is the *iff*. Nothing outside the class can be invariant, and everything inside it is invariant by construction—**no verification loop is required**, and no amount of engineering can rescue a statistic that lies outside.

Certificate. Invariance under a translation generator is equivalent to the directional derivative along it vanishing, so the claim is checkable with two central differences per marker.

4.3 The clinical vector, classified a priori

Seven of nine are exactly invariant *a priori* (Table 4). The final row is the useful kind of negative result: any marker keyed on absolute F_0 or absolute vocal-tract length is unpreservable by construction, because those are precisely the coordinates the mechanism releases. One should not engineer around it.

Table 4: A priori classification of the clinical markers under G.

Marker	Built from	Class
`f0_var_st`, `f0_range_st`	ψ	**exact**
`jitter_rel`	relative period perturbation (scale-free)	**exact**
`voiced_frac`, `pause_ratio`	F_0 presence / frame energy	**exact**
`shimmer_rel`, `loud_var_db`	frame energy	**exact**
`spec_var_db`, `artic_dyn_db_s`	log envelope (requires the Mellin half)	approximate
`f0_mean_st`, `a_log_vtl`	absolute F_0 / absolute VTL	**orbit coordinate:** destroyed by design

4.4 The design law

Theorem 2 (Dimension accounting)**.** $\dim(\textit{invariants}) = \dim(X) - \dim(G)$.

Every parameter added to the anonymization group costs exactly one functionally independent invariant dimension. This is the quantitative privacy–utility statement of the whole system, and it is why G is held at two parameters (Table 5).

Table 5: Consequences of enlarging the anonymization group.

Group	$\dim G$	Invariant class
$\{\alpha\} \times \{\beta\}$ — implemented	2	all prosodic shape, all amplitude, all timing: exact
+ F_0-range scaling	3	**loses F_0 variability**
+ 4-knot warp curve	6	loses most spectral-shape invariants

The middle row is a clinical-safety argument, not a utility one. F_0-range scaling is a *dilation* of ψ, not a translation, and F_0 variability is exactly the coordinate it quotients out. Since reduced F_0 variability is among the strongest acoustic correlates of depression [5], a system with that knob would cause a downstream classifier to read the output as **more depressed than the patient is**. We regard that as a defect of a different kind from a utility regression, and the system reports it under a dedicated `flattening` diagnostic.

4.5 Two implementation decisions that make the *iff* hold in practice

1. **A true dilation, not a power-law bin warp.** The warp $\nu' = \nu^w$ used by VTLN-P is *not a member of this group*, so none of the theory applies to it. ORBIT resamples the linear-frequency axis by $f \mapsto \alpha f$.
2. **The action is energy-preserving per frame.** A dilation on a finite band changes frame power, which would leak into the loudness, shimmer and pause markers and demote three exact invariants to approximate. Re-normalising frame energy costs nothing and buys them back.

4.6 The G_T sub-action and the timbre theorem

The group above moves the orbit coordinates. A second, larger action moves *timbre* at no invariant cost. Let $G_T = \{c \in \mathbb{R}^F\}$ act on the log envelope by a static additive curve followed by exact per-frame energy restoration.

Theorem 3 (G_T invariance)**.** *With* $g = \text{mean}_f\, L$, $S_h = L - g$, $\bar{m} = \text{mean}_t\, S_h$ *and* $D = S_h - \bar{m}$, *the action* $L \mapsto L + c$ *gives*

$$g \mapsto g + \underset{f}{\text{mean}}\, c, \qquad S_h \mapsto S_h + (c - \underset{f}{\text{mean}}\, c), \qquad \bar{m} \mapsto \bar{m} + (c - \underset{f}{\text{mean}}\, c), \tag{14}$$

and therefore

$$\boxed{D(t, f) \mapsto D(t, f) \quad \textit{identically.}} \tag{15}$$

Proof. $L \mapsto L + c$ gives $g \mapsto g + \text{mean}_f\, c$ directly. Then $S_h = L - g \mapsto L + c - g - \text{mean}_f\, c = S_h + (c - \text{mean}_f\, c)$. The added term is time-invariant, so averaging over t carries it into $\bar{m}$ unchanged. Subtracting, $D = S_h - \bar{m}$ maps to D exactly. Per-frame energy restoration fixes $E(t)$ by construction. □

Corollaries. `spec_var_db` and `artic_dyn_db_s` are functionals of D alone and are therefore invariant; `pause_ratio`, `shimmer_rel` and `loud_var_db` are functionals of $E(t)$ and are restored exactly; the ψ-markers never see the envelope. **All nine are invariant**, with measured worst-case drift $2.7 \times 10^{-14}\%$.

This is why timbre is the channel the mechanism attacks. G_T is F-dimensional and by Theorem 2 costs F invariant dimensions—but every one of them is a direction no clinical marker here reads, while the long-term average spectrum is exactly what a speaker-verification embedding leans on hardest. **Identity can be moved a long way at no measured clinical cost**, and PolarQuant (§ 3.2) is the mechanism that moves it.

4.7 Where the theory stops

Stated explicitly, because each boundary was found by measurement:

- **Mellin is exact only on the infinite log-frequency line.** On a finite band a dilation pushes content past the edges. Guard bands, windowing and a tighter band were each tried and each made it worse. So: **the pitch subgroup admits an exact maximal invariant; the VTL subgroup does not.**
- **The vocoder floor, not the theory, is the end-to-end error floor.** In the parameter domain the exact invariants drift by 0.000%. Re-analysing a *synthesized* waveform re-estimates F_0 and the envelope and reintroduces several percent. Clinical statistics are therefore computed on the WORLD parameter track, and the analysis–synthesis floor is measured separately and subtracted as `mechanism_excess_pct`.
- **A downstream classifier need not factor through (ψ, M).** If it reads the orbit coordinates, no invariance theory covers it, and the system provides a flip-rate test against the real model rather than assuming the proxy transfers.

4.8 Privacy accounting in two separate halves

Orbit coordinate — μ-GDP. As in § 3.3. The knob is the **shrinkage, not the noise**: $\mu \to 0$ forces $\lambda \to 0$ and the released coordinate becomes independent of the speaker. μ is *distributional*—it bounds how distinguishable the input is, and does not promise any single draw lands far from the speaker. A draw may legitimately land near identity (measured: d-vector cosine 0.96 on one such draw). Forcing a minimum displacement is **not** differential privacy, because it conditions the output on the input's distance; it is offered and flagged `gdp_exact: false` wherever it fires.

Clinical channel — k-bit cap. The seven exact invariants are never re-imposed, so they open no channel. The two approximate markers are: re-imposing a measured value writes it into the output. A **dithered** k-bit quantizer caps that at k bits per repaired marker. Dither is not cosmetic—without it the quantizer grid is learnable by any model later trained on the corpus. The repair is skipped, with the reason recorded, whenever the quantizer is coarser than the drift it would correct.

Impossibility, stated up front. Any mechanism preserving a clinical vector C exactly leaks at least $I(\text{identity}; C)$ bits. Perfect clinical preservation and perfect anonymization are **jointly impossible** whenever $I(S; C) > 0$. The contribution is not beating that floor; it is decomposing the budget and reporting both halves.

4.9 Processing chain and algorithm

Table 6 summarises the processing chain; Algorithm 1 gives the per-speaker release.

Table 6: ORBIT-N+F processing chain.

Analysis	WORLD Harvest → StoneMask → CheapTrick → D4C, 5 ms frames, segmented at 120 s. Speaker frames from the transcript, nearest-filled. Per speaker: mean shape $\bar{m}$, cepstra 1..48, centroid a_{in}, mean ϕ; clinical reference $C(x)$.
Stage A (timbre)	$\tau_{\text{in}} = \text{to_tau}(c_{\text{in}}, \phi)$ [keyed: $\ldots \| \text{polar}$]; Haar rotation → keep b sign bits ($b = 0$) → resample direction → fresh χ_K radius, q_{95}-projected → τ_{out}. Curve $c(f) = \text{IDCT}(\text{from_tau}(\tau_{\text{out}}) - c_{\text{ref}})$, with c_{ref} measured on the *dilated* shape because the renderer applies the dilation first.
Stage B (formant)	$a_{\text{out}} = \text{anchor} + \lambda\ \text{clip}(a_{\text{in}}) + \sigma z$ [keyed: $\ldots \|$session] → output-box projection (μ-safe) → shift, capped ±4 st (flagged). Public anchors: 289 Hz, $\Delta = 0.324$.
Stage C (render)	mel of the original at 24 kHz → liftered envelope dilation (20 coefficients, tapered top octave) → + timbre curve on 80 bands → per-frame energy restored (G_T completes here) → neural decode with phase-continuous excitation → overlap-add (20 s windows, 1 s crossfade) → resample → RMS match.
Verify	Re-analyse the output; per-speaker end-to-end drift against $C(x)$; d-vector cosine per speaker; compare against the renderer floor.

Two orderings in this algorithm are load-bearing and were both found by measurement. **Line 6:** the reference shape for the timbre curve must be the *dilated* shape, because the renderer applies the dilation first; measuring against the undilated shape overshoots into the high band by ∼7 dB on the largest shift and inflates apparent privacy. **Line 11:** energy restoration must follow the additive curve, or three exact invariants are demoted to approximate.

4.10 Why a neural decoder, and which one

Three measured results determined the renderer, and they are not interchangeable:

1. **Moving formants *in time* destroys intelligibility.** A time-varying warp field cost 14 CER points and bought no privacy; removing it cost nothing. Formants carry vowel identity, so a *consistent* vocal-tract change is acceptable and a *moving* one is not. Hence the dilation is static per speaker.
2. **A neural decoder helps only if a DSP vocoder is not upstream of it.** WORLD → mel → neural measured no better than WORLD alone. The neural decoder on the *original's*

Algorithm 1 Per-speaker release

Require: speech frames for speaker s, public space P, secret K, session σ

$\bar{m} \leftarrow \text{mean}_t\,(L - \text{mean}_f\, L)$ over s's speech frames
$c \leftarrow \text{DCT-II}(\bar{m})[1..48]$; $\bar{\phi} \leftarrow$ mean log-F_0 of s
$a_{\text{in}} \leftarrow \textsc{ShapeCentroid}(\text{IDCT}(c))$ ▷ on the public-space grid
$rng_B \leftarrow \textsc{Seed}(K, \text{item}, s, \sigma)$; $rel \leftarrow \mu$-GDP release of $(a_{\text{in}}, \bar{\phi})$
$a_{\text{shift}} \leftarrow rel.a_{\text{shift}}$; flag `gdp_exact`
$shape_{\text{ref}} \leftarrow \textsc{Dilate}(\bar{m}, a_{\text{shift}})$ ▷ renderer applies dilation first
$rng_A \leftarrow \textsc{Seed}(K, \text{item}, s, \sigma \,\|\, \text{“polar”})$
$\tau_{\text{out}} \leftarrow \textsc{PolarQuant}(\text{to_tau}(c, \bar{\phi}), rng_A, b)$
$curve \leftarrow \text{IDCT}\big(\text{from_tau}(\tau_{\text{out}}, \bar{\phi}) - \text{DCT-II}(shape_{\text{ref}})[1..48]\big)$
$mel \leftarrow \textsc{LifteredDilate}(mel_s, a_{\text{shift}}) + \textsc{OnBands}_{80}(curve)$
$mel \leftarrow \textsc{RestoreFrameEnergy}(mel)$ ▷ G_T completes
return $\textsc{Decode}(mel)$, public report, private sidecar

mel measured markedly better, because a mel computed from the real waveform carries cycle-level detail that a 5 ms F_0 track cannot.

3. **Excitation must be phase-continuous across chunks.** The stock harmonic-plus-noise source restarts phase on every call, which places a click at every chunk boundary and makes two crossfaded chunks carry different excitations.

4.11 Benchmarks for ORBIT-N+F itself

Table 7 gives the ablation across the design history, with the same corpus and evaluator throughout.

Table 7: Ablation across the design history. EER and self-cosine in %.

Variant	What changed	EER ↑	self-cos ↓	CER ↓
VTLN-P jitter	baseline, not in the group	24.7	39.8	18.5
ORBIT-3c	μ-GDP release, DSP render, warp field	28.0	40.2	33.6
ORBIT-4	**+ PolarQuant timbre replacement**	35.2	27.9	32.9
ORBIT-4n	− warp field	35.7	26.6	**18.8**
ORBIT-N+F	neural render on the original's mel	32.6 ± 1.2	27.8 ± 4.6	**14.6**

Three transitions carry the whole result. **3c → 4** is Theorem 3 in action: replacing timbre rather than perturbing the orbit coordinates takes self-cosine from 40.2% to 27.9%, because G_T moves the long-term spectrum—what the embedding reads—at no invariant cost. **4 → 4n** removes the warp field for a 14-point CER gain at no privacy cost, confirming result (1) of § 4.10: the identity work had already moved to a channel that costs nothing. **4n → N+F** is the renderer alone, 18.8% → 14.6% CER with privacy unchanged, confirming result (2).

Clinical performance, end-to-end. Table 8 reports relative marker drift for the 13 speakers with ≥ 30 s of speech (two shorter speakers excluded as unstable).

The identity-floor column is essential: rendering the **un-anonymized** file through the same decoder already moves F_0 variability by 3.4% on average. Much of the residual is the renderer, not the mechanism—and for four markers the floor equals or exceeds the measured drift.

Pitch preservation. Across all 15 speakers and full files, the median |shift| is **0.02 st** and the maximum **0.08 st**, including a speaker given a +8.2 st formant shift. The liftering keeps harmonics in place.

Table 8: End-to-end relative drift of clinical markers for ORBIT-N+F (13 speakers). The final column is the drift incurred by rendering the *un-anonymized* file through the same decoder.

Marker	mean	median	worst	renderer identity floor
`f0_var_st`	5.3%	2.0%	17.1%	3.4%
`f0_range_st`	5.6%	4.7%	16.3%	3.0%
`jitter_rel`	4.1%	2.2%	13.1%	4.5%
`voiced_frac`	4.8%	3.9%	9.2%	3.8%
`pause_ratio`	7.1%	3.1%	33.7%	2.4%
`shimmer_rel`	5.0%	4.8%	10.3%	2.4%
`loud_var_db`	1.5%	1.2%	4.8%	0.5%
`spec_var_db`	3.0%	3.2%	5.1%	4.3%
`artic_dyn_db_s`	4.0%	4.8%	7.0%	6.1%

4.12 Reporting discipline

Two conventions are part of the contribution, because both were adopted after a measurement went wrong without them.

Report the CMF as a triple, never a scalar: (i) the per-marker drift vector, unaggregated, because a mean hides the single failure that matters; (ii) Grassmann tangent alignment, where high is good on the *clinical* manifold and **low is the success signal** on the *spectral* manifold—the two numbers must never be compared; (iii) decision margin on the operator's real classifier, reported as `NOT MEASURED` unless one is supplied, never defaulted.

Always read the per-speaker block. On a two-party consultation, whole-file markers mix clinician and patient. Most F_0 variance is *between* them, so a mechanism that legitimately moves the two speakers toward each other appears as a $\sim$40% collapse in "F_0 variability" that neither speaker experienced.

5 Reversibility and security analysis

Why classical DSP is invertible. McAdams warping is $\theta \mapsto \theta^{\alpha}$ on LPC pole angles—a bijection parameterised by a scalar in roughly $[0.8, 1.2]$. An attacker sweeps α, inverts, and scores; identity entropy $H(\text{Identity})$ is conserved because the map is one-to-one. The same holds for a uniform VTLN warp. Our measurements are consistent: 57.6% and 39.8% self-cosine respectively.

Why the present construction is not invertible in the same sense. Three distinct barriers compose:

1. **Information-theoretic erasure.** At $b = 0$ the released timbre is statistically independent of the input (Proposition 1). There is no residual to recover because there is no dependence—inversion is not hard, it is undefined.
2. **Many-to-one statistical mapping.** The μ-GDP release maps a clipped input range onto a noised output, so the posterior over a_{in} given a_{out} is the prior reweighted by a bounded likelihood ratio. MLE recovers the anchor, not the speaker, as $\lambda \to 0$ ($\lambda_a = 0.054$ here).
3. **Secret rotation.** R is drawn Haar on $\mathbb{S}^{K-1}$ from a keyed stream; without the secret the orientation is unknown.

Table 9: Privacy and intelligibility. EER, self-cosine and CER in %. "—" indicates not computed for that configuration.

Method	Family	EER ↑	self-cos ↓	π_{link} ↓	min C_{llr} ↑	CER ↓
Original (untouched)	—	9.8	100	—	—	0
McAdams $\alpha = 0.92$	DSP	17.3	57.6	—	—	14.3
VTLN-P w1.1/+2 st	DSP	19.8	44.8	—	—	—
VTLN-P jitter	DSP	24.7	39.8	—	—	18.5
ORBIT-3c	DSP + μ-GDP	28.0	40.2	—	—	33.6
ORBIT-4	+ PolarQuant	35.2	27.9	—	—	32.9
ORBIT-4n	+ PolarQuant	35.7	26.6	—	—	18.8
ORBIT-N+F (HiFT)	neural render	32.6 ± 1.2	27.8 ± 4.6	—	—	**14.6**
OpenVoice V2 only	*control*	14.7	64.5	**1.000**	0.143	15.4
Single-pass → OpenVoice	neural back end	35.4	2.6	0.214	0.725	31.1
ORBIT-N+F → OpenVoice	neural back end	**40.3**	**−4.3**	**0.071**	**0.799**	35.5

We deliberately do **not** describe this as cryptographic. μ-GDP is a statistical guarantee with a stated parameter, not a hardness assumption, and the composition across recordings is untracked (§ 8).

6 Experimental results

6.1 Privacy and intelligibility

All rows in Table 9 use 14 identities, 851 chunks and one evaluator. CER is the mean over the three files every configuration shares.

For the final row, $\pi_{\text{sing}} = 0.0$ ($p = 5.6 \times 10^{-140}$ against a 0.368 baseline) and the ROCCH EER is 40.7%.

The privacy is the mechanism's. The control row leaves $\pi_{\text{link}} = 1.000$—every speaker still correctly linked after a converter round trip. Whatever the full pipeline achieves, the converter did not do it.

Privacy rises monotonically with the mechanism's strength, and the decisive step is ORBIT-3c → ORBIT-4: replacing timbre rather than perturbing it takes self-cosine from 40.2% to 27.9%.

6.2 Non-parametric back end

A second back end was evaluated in which the source is rebuilt from the nearest WavLM [3] frames of a target speaker, following kNN-VC [1], with the target drawn from a 16-speaker public RixVox pool (validated: within-label cosine 0.332 vs. between-label −0.022, margin 0.354). Table 10 reports results on one recording; STOI follows Taal et al. [19].

Table 10: Non-parametric (kNN-VC) back end, single recording. CER, WER and self-cosine in %.

	CER	WER	STOI	self-cos
Private kNN-VC	31.8	55.2	0.625	−10.8
ORBIT-N+F → private kNN-VC	42.5	67.8	0.610	−15.4

Chaining costs 10.7 CER points for ∼5 points of self-cosine already past zero. These rows are single-file and carry **no** EER or π_{link}, which are corpus statistics over identities.

6.3 Clinical markers, end-to-end

Table 11 reports ICC(2,1) [18] between original and release over 15 speaker instances.

Table 11: ICC(2,1) between original and release, 15 speaker instances.

Marker	OpenVoice only (no anonymization)	ORBIT-N+F → OpenVoice
`loud_var_db`	0.989	**0.309**
`pause_ratio`	0.978	**0.061**
`f0_var_st`	0.916	**0.172**
`f0_range_st`	0.865	**0.270**
`spec_var_db`	0.832	**0.218**
`artic_dyn_db_s`	0.575	**−0.058**
`jitter_rel`	0.556	**−0.321**
`voiced_frac`	0.421	**0.122**
`shimmer_rel`	0.337	**0.021**

Zero of nine markers clear the 0.75 threshold [11]. The parameter-domain theorem of § 3.4 holds; it does not survive the decoder. We report this rather than quote the parameter-domain figure as a system property.

6.4 Artifact audit

A verification embedding cannot distinguish a different speaker from a broken signal, so an EER gain is only evidence of anonymization if the signal is intact. We test the specific failure modes this architecture could produce (Table 12).

Table 12: Artifact audit: control (OpenVoice only) vs. full pipeline.

	control	full pipeline
bandwidth retained (99% energy)	93.3%	164.3%
worst octave-band temporal SD ratio	0.85	1.40
splice-seam flux ÷ interior flux	×1.01	×1.07
STOI	0.821	0.692

Nothing is frozen (a pinned band reads ∼0) and per-speaker seams are indistinguishable from interior points. The release is wider-band than the source because the clinical recordings carry 99% of their energy below 1.6–2.8 kHz and the decoder synthesises a band the input never had.

7 Negative results

These constrain the design space more than the positive results do, and each contradicts a hypothesis we held.

7.1 The stages are additive, not compounding

We hypothesised that cascading a spectral pre-warp into a converter causes compounding distortion. Table 13 reports a 2×2 over the two stages, every cell on the same two files.

Interaction: −1.2 points. Compounding distortion predicts a *positive* interaction. The measured sign is the opposite. Replacing the timbre costs roughly three times what the dilation costs, and they do not compound. Across 30 further clips the applied formant shift showed **no significant correlation** with CER (Pearson $r = 0.151$, $p = 0.43$), with a fitted intercept of 22.1% CER at zero shift.

Table 13: 2×2 factorial over Stage A and Stage C. Cells are CER.

	Stage C off	Stage C on	Stage C effect
Stage A off	18.6%	23.9%	+5.2
Stage A on	34.2%	38.3%	+4.1
Stage A effect	**+15.5**	**+14.4**	

7.2 The embedding-regime hypothesis does not explain the cost

We observed that public pool embeddings sit at whitened radius ∼9.5 while embeddings extracted from real recordings sit at 28–34, and hypothesised that this regime gap pushes the flow off-distribution. A controlled swap of the target to a same-regime public voice, with the dilation unchanged, recovered 11.7 of 21.8 CER points on one file—supporting the hypothesis.

It does not survive a second corpus. On 30 VoxLingua [21] Swedish clips the source geometry is **identical** to the clinical corpus ($r_{\text{in}} = 29.95 \pm 0.68$ vs. 29.29 ± 2.84; target-distance proxy 38.8 vs. 38.9) yet CER differs by 11.5 points (22.6% vs. 34.1%). Distance alone does not predict the damage. A non-parametric back end with no conditioning vector at all (§ 6.2) nonetheless costs 31.8% CER, which points to the conclusion that **the cost is intrinsic to displacing a voice this far**, not to how the target is represented.

7.3 Measured intelligibility is strongly recognizer-dependent

The benchmark uses a CTC recognizer for comparability with published baselines. Spot checks with stronger systems reported 1.9–5.4× lower error on identical audio (voxrex 5.5% → 0%; 9.6% → 1.79%; 50.9% → 27.5%).

The generalisation fails at corpus scale. Re-running all 30 VoxLingua clips with `kb-whisper-large` [10] gives Table 14.

Table 14: Recognizer dependence on 30 VoxLingua Swedish clips.

	voxrex	kb-whisper-large
CER median	22.6%	**38.9%**
WER median	40.8%	**47.5%**

Error went **up**, and two clips exceeded 100% CER—the hypothesis far longer than the reference, the signature of a generative decoder looping on off-distribution acoustics. A CTC model cannot fail this way; its output length is bounded by the input. **A stronger recognizer is not uniformly a more robust one on anonymized speech**, and absolute intelligibility figures should be read as recognizer-conditional.

7.4 A baseline is required for a privacy number to mean anything

On the VoxLingua corpus the pipeline scores EER 50.4% and self-cosine −3.0%, which reads as excellent. The baseline EER on **unmodified** audio is **34.3%**—near chance—because each clip concatenates 2–5 speakers with no labels, so enrol and test halves often compare different people. With almost no identity present to remove, the 50.4% measures nothing about the anonymizer. The CER of 22.6% from that corpus is valid; the EER is not.

8 Limitations

- **Pitch is not anonymized.** No stage moves F_0; register and probable gender survive. This is deliberate, and stated.

- **The clinical invariance is parameter-domain.** End-to-end, with a generative decoder downstream, zero of nine markers clear ICC 0.75.
- **Single session draw** for the neural rows. The predecessor's published figure carries ±1.2 EER points over three draws; smaller differences are not signal.
- **The ±4 st cap is not post-processing.** It did not fire here (15/15 exact), but it voids the exact bound when it does.
- **No cross-recording composition.** N recordings of one speaker compose as $\approx\sqrt{N}\mu$; this is untracked.
- **Content is not redacted.** Names and identifiers in the *words* survive every configuration reported.
- **Attribution rests on two files** for the 2×2. The main effects far exceed plausible noise; the -1.2 interaction does not, and should be read as "no interaction detected."
- **No external benchmark.** All comparisons are internal, on one corpus, with one embedder. A claim of state of the art would require a standard protocol such as the VoicePrivacy Challenge [20], which we have not run.

9 On legal framing

$\pi_{\text{link}} = 0.071$ equals $1/N$ for $N = 14$, and $\pi_{\text{sing}} = 0.0$ at $p = 5.6 \times 10^{-140}$. These are strong evidence against linkability and singling-out **for this corpus, this attacker and this embedder**, and they are the quantities a data-protection assessment under Recital 26 of the GDPR [9] would weigh.

They are not a compliance determination, and this paper does not make one. Three gaps are material to any such assessment: content is unredacted; cross-recording composition is untracked; and the attacker model is one embedder, where a determination would want several, including models retrained on anonymized speech.

10 Conclusion

We presented a timbre-replacement mechanism with a zero-bit information cap, a μ-GDP release of vocal-tract scale against public anchors, and a pitch-preserving envelope dilation, and measured what they achieve with a neural back end.

The headline is genuine: **π_{link} at chance, $\pi_{\text{sing}} = 0$, self-cosine -4.3%, and a control proving the privacy is the mechanism's and not the vocoder's.** So is the cost: **CER 34.1% against 14.6%, and zero of nine clinical markers above ICC 0.75.**

We therefore do not claim to resolve the privacy–utility–fidelity trilemma. We claim to have characterized it, and three negative results sharpen where the boundary lies: the stages are additive rather than compounding; the cost is intrinsic to the displacement rather than to the target's representation; and measured intelligibility is recognizer-conditional in a way that can invert with a stronger model.

The operational consequence is specific. For releases where the words are the payload, the configuration reported here provides the strongest unlinkability we measured. **For clinical corpora whose value is the biomarkers, ORBIT-N+F's own renderer remains the better system**—14.6% CER at 27.8% self-cosine—and the neural back end should not be adopted for that purpose on this evidence.

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